

PRACTICE SHEET 1

1. A man buys a cycle for 1400 and sells it at a loss of 15%. What is the selling price of the cycle?
 (1) 1202 (2) 1190
 (3) 1160 (4) 1000
2. On selling an article for 651, there is a loss of 7%. The cost price of that article is
 (1) 744 (2) 751
 (3) 793 (4) 700
3. A milkman bought 70 litres of milk for 630 and added 5 litres of water. If he sells it at 9.00 per litre, his profit percentage is
 (1) $8\frac{1}{5}\%$ (2) 7%
 (3) $8\frac{2}{5}\%$ (4) $7\frac{1}{7}\%$
4. In terms of percentage profit, which is the best transaction?

C.P. (in)	Profit (In)
(I) 36	17
(II) 50	24
(III) 40	19
(IV) 60	29
(1) I	(2) II
(3) III	(4) IV
5. A man bought an old typewriter for 1200 and spent 200 on its repair. He sold it for 1680. His profit per cent is:
 (1) 20% (2) 10%
 (3) 8% (4) 16%
6. If the cost price is 95% of the selling price, what is the profit percent?
 (1) 4% (2) 4.75%
 (3) 5% (4) 5.26%
7. A merchant buys an article for 27 and sells it at a profit of 10% of the selling price. The selling price of the article is :
 (1) 29.70 (2) 30
 (3) 37 (4) 32
8. If the cost price of an article is 80% of its selling price, the profit per cent is :
 (1) 20 % (2) $22\frac{1}{2}\%$
- (3) 24% (4) 25%
9. Krishnan bought a camera and paid 20% less than its original price. He sold it at 40% profit on the price he had paid. The percentage of profit earned by Krishnan on the original price was
 (1) 22% (2) 32%
 (3) 12% (4) 15%
10. By selling a car for 64,000, Mr. Rao lost 20%. Then the cost price of the car is :
 (1) 72,000 (2) 76,800
 (3) 80,000 (4) 84,000
11. By selling an article, a man makes a profit of 25% of its selling price. His profit per cent is
 (1) 20% (2) 25%
 (3) $16\frac{2}{3}\%$ (4) $33\frac{1}{3}\%$
12. If there is a profit of 20% on the cost price of an article, the percentage of profit calculated on its selling price will be
 (1) 24 (2) $16\frac{2}{3}$
 (3) $8\frac{1}{3}$ (4) 20
13. A man purchased a bed sheet for Rs 450 and sold it at a gain of 10% calculated on the selling price. The selling price of the bed sheet was
 (1) Rs 460 (2) Rs 475
 (3) Rs 480 (4) Rs 500
14. By selling an article for 960 a man incurs a loss of 4%; what was the cost price?
 (1) 1,000 (2) 784
 (3) 498.4 (4) 300
15. A salesman expects a gain of 13% on his cost price. If in a month his sale was 7,91,000, what was his profit ?
 (1) 85,659 (2) 88,300
 (3) 91,000 (4) 97,786
16. A merchant purchases a wrist watch for 450 and fixes its list price in such a way that after allowing a discount of 10%, he earns a

- profit of 20%. Then the list price of the watch is
 (1) 650 (2) 700
 (3) 550 (4) 600
- 17.** The cost price of a radio is 600. The 5% of the cost price is charged towards transportation. After adding that, if the net profit to be made is 15%, then the selling price of the radio must be
 (1) 704.50 (2) 724.50
 (3) 664.50 (4) 684.50
- 18.** If there is a profit of 20% on the cost price, the percentage of profit on the sale price is
 (1) $16\frac{2}{3}\%$ (2) 12 %
 (3) $15\frac{1}{3}\%$ (4) 16 %
- 19.** Nisha bought a number of oranges at 2 for a rupee and an equal number at 3 for a rupee. To make a profit of 20% she should sell a dozen for
 (1) 6 (2) 8
 (3) 10 (4) 12
- 20.** If selling price of an article is $1\frac{1}{3}$ of cost price, find gain %.
 (1) 25% (2) $33\frac{1}{3}\%$
 (3) 1.33% (4) $66\frac{2}{3}\%$
- 21.** A merchant loses 10% by selling an article. If the cost price of the article is 15, then the selling price of the article is
 (1) 13.20 (2) 16.50
 (3) 12.30 (4) 13.50
- 22.** Pooja wants to sell a watch at a profit of 20%. She bought it at 10% less and sold it at 30 less, but still she gained 20%. The cost price of watch is
 (1) 240 (2) 220
 (3) 250 (4) 225
- 23.** A fruit merchant makes a profit of 25% by selling mangoes at a certain price. If he charges Rs.1 more on each mango, he would gain 50%. At first the price of one mango was
 (1) Rs. 5 (2) Rs. 7
 (3) Rs. 4 (4) Rs. 6
- 24.** There is 10% loss if an article is sold at Rs. 270. Then the cost price of the article is
 (1) Rs. 300 (2) Rs. 270
 (3) Rs. 320 (4) Rs. 250
- 25.** If bananas are bought at the rate of 4 for a rupee, how many must be sold for a rupee so as to gain $33\frac{1}{3}\%$?
 (1) 2.5 (2) 2
 (3) 3 (4) 4
- 26.** If the profit on selling an article for Rs. 425 is the same as the loss on selling it for Rs. 355, then the cost price of the article is
 (1) Rs. 410 (2) Rs. 380
 (3) Rs. 400 (4) Rs. 390
- 27.** The C.P of 10 articles is equal to the S.P. of 15 articles. What is the profit or loss percentage?
 (1) 25.5% (2) 35%
 (3) 10% (4) 33.3%
- 28.** The selling price of 6 bananas is equal to the cost price of 8 bananas. Then the percentage of profit is:
 (1) 20 (2) $33\frac{1}{3}$
 (3) 25 (4) 30
- 29.** By selling a bag at Rs. 230, profit of 15% is made. The selling price of the bag, when it is sold at 20% profit would be
 (1) Rs. 250 (2) Rs. 205
 (3) Rs. 240 (4) Rs. 200
- 30.** A man gains 20% by selling an article for a certain price. If he sells it at double the price, the percentage of profit will be
 (1) 40% (2) 100%
 (3) 120% (4) 140%

ANSWER KEY

1.	(2)	2.	(4)	3.	(4)	4.	(4)	5.	(1)	6.	(4)	7.	(2)	8.	(4)	9.	(3)	10.	(3)
11.	(4)	12.	(2)	13.	(4)	14.	(1)	15.	(3)	16.	(4)	17.	(2)	18.	(1)	19.	(1)	20.	(2)
21.	(4)	22.	(3)	23.	(1)	24.	(1)	25.	(3)	26.	(4)	27.	(4)	28.	(2)	29.	(3)	30.	(4)

PRACTICE SHEET 2

- The cost price of 36 books is equal to the selling price of 30 books. The gain per cent is :
 (1) 20% (2) $16\frac{2}{3}\%$
 (3) 18% (4) $82\frac{1}{3}\%$
- The cost price of 15 articles is same as the selling price of 10 articles. The profit percent is:
 (1) 30% (2) 40%
 (3) 50% (4) 45%
- The selling price of 5 articles is the same as the cost price of 3 articles. The gain or loss percent is:
 (1) 20% gain (2) 25% gain
 (3) 33.33% loss (4) 40% loss
- If the cost price of 15 tables be equal to the selling price of 20 tables, the loss per cent is:
 (1) 20% (2) 30%
 (3) 25% (4) 37.5%
- The cost price of 18 articles is equal to the selling price of 15 articles. The gain per cent is
 (1) 15% (2) 20%
 (3) 25% (4) 18%
- A person sells 400 mangoes at the cost price of 320 mangoes. His percentage of loss is
 (1) 10% (2) 15%
 (3) 20% (4) 25%
- If the cost price of 50 oranges is equal to the selling price of 40 oranges, then the profit per cent is
 (1) 5% (2) 10%
 (3) 20% (4) 25%
- If the cost price of 12 oranges is equal to selling price of 10 oranges, then the percentage of profit is
 (1) $16\frac{2}{3}\%$ (2) 20%
 (3) 18% (4) 25%
- If the cost price of 10 articles is equal to the selling price of 9 articles, the gain or loss per cent is
 (1) $11\frac{1}{9}\%$ profit (2) $7\frac{6}{17}\%$ profit
 (3) $11\frac{1}{9}\%$ loss (4) $1\frac{12}{13}\%$ loss
- A man sells 320 mangoes at the cost price of 400 mangoes. His gain percent is:
 (1) 15% (2) 20%
 (3) 25% (4) 10%
- If the cost price of 12 pens is equal to the selling price of 8 pens, the gain per cent is :
 (1) $33\frac{1}{3}\%$ (2) $66\frac{2}{3}\%$
 (3) 25% (4) 50%
- The cost price of 8 articles is equal to the selling price of 9 articles. The profit or loss per cent in the transaction is
 (1) $12\frac{1}{2}\%$ loss (2) $12\frac{1}{2}\%$ profit
 (3) $11\frac{1}{9}\%$ loss (4) $11\frac{1}{9}\%$ profit
- A sold an article to B at 20% profit and B sold to C at 15% loss. If A sold it to C at the selling price of B, then A would make

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| <p>(1) 5% profit (2) 2% profit
(3) 2% loss (4) 5% loss</p> <p>14. If the cost price of 10 articles is equal to the selling price of 7 articles, then the gain or loss per cent is:</p> <p>(1) 51% gain (2) $42\frac{6}{7}$ % gain
(3) 35 % loss (4) $42\frac{6}{7}$ % loss</p> <p>15. Mahesh purchased a radio at $\frac{9}{10}$ of its selling price and sold it at 8% more than its original selling price. His gain per cent is :</p> <p>(1) 20 % (2) 18%
(3) 10 % (4) 8 %</p> <p>16. A coconut merchant finds that the cost price of 2750 coconuts is the same as the selling price of 2500 coconuts. The loss or gain per cent is</p> <p>(1) 5% loss (2) 15% loss
(3) 20% gain (4) 10% gain</p> <p>17. If the cost price of 10 articles is equal to the selling price of 16 articles, then the loss per cent is</p> <p>(1) 30% (2) 37.5%
(3) 42.5% (4) 45%</p> <p>18. If the selling price of 4 articles is equal to the cost price of 5 articles, the profit percent is</p> <p>(1) 20% (2) $22\frac{1}{2}$ %
(3) 25% (4) 30%</p> <p>19. The selling price of 10 oranges is the cost price of 13 oranges. Then the profit percentage is</p> <p>(1) 30% (2) 10%
(3) 13% (4) 3%</p> <p>20. If the selling price of 10 articles is equal to the cost price of 11 articles, then the gain percent is</p> <p>(1) 10% (2) 11%
(3) 15% (4) 25%</p> <p>21. A cloth merchant on selling 33 metres of cloth obtains a profit equal to the selling price of 11 metres of cloth. The profit percent is</p> | <p>(1) 40% (2) 22%
(3) 50% (4) 11%</p> <p>22. A shopkeeper buys 144 items at 90 paise each. On the way 20 items are broken. He sells the remainder at 1.20 each. His gain per cent correct to one place of decimal is</p> <p>(1) 13.8% (2) 14.6%
(3) 14.8% (4) 15.8%</p> <p>23. If goods be purchased for 450 and one third sold at a loss of 10%. At what gain percent should the remainder be sold so as to gain 20% on the whole transaction?</p> <p>(1) 32% (2) 35%
(3) 28% (4) 30%</p> <p>24. A shoe company sold 50 pairs of shoes on a day costing 189.50 each for 10,000. Then the profit obtained in is</p> <p>(1) 522 (2) 525
(3) 573 (4) 612</p> <p>25. The cost price of 25 books is equal to the selling price of 20 books. The profit per cent is</p> <p>(1) 20% (2) 22%
(3) 24% (4) 25%</p> <p>26. If the selling price of 40 articles is equal to the cost price of 50 articles, the loss or gain per cent is</p> <p>(1) 25% gain (2) 20% gain
(3) 25% loss (4) 20% loss</p> <p>27. If the cost price of 20 books is the same as selling price of 25 books, then the loss percentage is</p> <p>(1) 20 (2) 25
(3) 22 (4) 24</p> <p>28. By what fraction selling price (S.P.) must be multiplied to get the cost price (C.P.) if the loss is 20%?</p> <p>(1) $\frac{4}{5}$ (2) $\frac{8}{5}$
(3) $\frac{5}{4}$ (4) $\frac{6}{5}$</p> <p>29. A store sells a watch for a profit of 25% of its cost price. Then the percentage of profit against selling price is:</p> <p>(1) 22% (2) 20%
(3) 18% (4) 15%</p> <p>30. The per cent profit made when an article is sold for Rs. 78 is twice as much as when it</p> |
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is sold for Rs. 69. The cost price of the article is

- (1) Rs. 60
(2) Rs. 51
(3) Rs. 55.50
(4) Rs. 70

ANSWER KEY

1.	(1)	2.	(3)	3.	(4)	4.	(3)	5.	(2)	6.	(3)	7.	(4)	8.	(2)	9.	(1)	10.	(3)
11.	(4)	12.	(3)	13.	(2)	14.	(2)	15.	(1)	16.	(4)	17.	(2)	18.	(3)	19.	(1)	20.	(1)
21.	(3)	22.	(3)	23.	(2)	24.	(2)	25.	(4)	26.	(1)	27.	(1)	28.	(3)	29.	(2)	30.	(1)

PRACTICE SHEET 3

- A table is sold at a profit of 13%. If it is sold for 25 more, profit is 18 %. Cost price of table is
(1) 100 (2) 500
(3) 200 (4) 1, 000
- A man sold his watch at a loss of 5%. Had he sold it for 56.25 more, he would have gained 10%. What is the cost price of the watch (in)?
(1) 370 (2) 365
(3) 375 (4) 390
- Kamala bought a bicycle for 1,650. She had to sell it at a loss of 8%. She sold it for
(1) 1, 581 (2) 1, 518
(3) 1, 510 (4) 1, 508
- A table is sold at Rs. 1,800 at a loss of 10%. At what price should it be sold to earn a profit of 15%?
(1) 2,070 (2) 1,890
(3) 2,000 (4) 2,300
- A manufacturer sells an item to a wholesale dealer at a profit of 18%. The wholesaler sells the same to a retailer at a profit of 20%. The retailer in turn sells it to a customer for 15045 thereby earning a profit of 25%. The cost price of the manufacturer is
(1) 8000 (2) 8500
(3) 9000 (4) 10000
- A man sold an article at a gain of 5%. Had he sold it for Rs. 240 more, he would have gained 8%. The cost price of the article is
(1) Rs. 6,000 (2) Rs. 10,000
(3) Rs. 12,000 (4) Rs. 8,000
- A radio is sold at a profit of 20%. Had it been sold for Rs. 60 more the profit would have been 30%. The cost price of the radio is
(1) Rs. 500 (2) Rs. 600
(3) Rs. 550 (4) Rs. 620
- A dealer sold a bicycle at a profit of 10%. Had he bought the bicycle at 10% less price and sold it at a price Rs. 60 more, he would have gained 25%. The cost price of the bicycle was
(1) Rs. 2400 (2) Rs. 2600
(3) Rs. 2000 (4) Rs. 2200
- If 3 articles are sold for the cost of 5 articles, then the profit percentage is :
(1) 50 (2) 60
(3) $66\frac{2}{3}$ (4) 65
- A sold a watch at a gain of 5% to B and B sold it to C at a gain of 4%. If C paid Rs. 91 for it, the price paid by A is:
(1) Rs. 83.33 (2) Rs. 84.33
(3) Rs. 83 (4) Rs. 82.81
- Arun buys one kilogram of apples for Rs. 120 and sells it to Swati gaining 25%. Swati sells it to Divya who again sells it for Rs. 198, making a profit of 10%. What is the profit percentage made by Swati?
(1) 25% (2) 20%
(3) 16.67% (4) 15%
- A dealer sold an article at 6% loss. Had he sold it for Rs. 64 more, he would have made a profit of 10%. Then the cost of the article is
(1) Rs. 400 (2) Rs. 200
(3) Rs. 164 (4) Rs. 464
- Profit after selling a commodity for 524 is the same as loss after selling it for 452. The cost price of the commodity is
(1) 480 (2) 500

- (3) 488 (4) 485
- 14.** A clock was sold for 144. If the percentage of profit was numerically equal to the cost price, the cost of the clock was
 (1) 72 (2) 80
 (3) 90 (4) 100
- 15.** By selling 144 hens Mahesh suffered a loss equal to the selling price of 6 hens. His loss percent is
 (1) 4% (2) 3%
 (3) 9% (4) $4\frac{1}{2}\%$
- 16.** If the profit per cent got on selling an article is numerically equal to its cost price in rupees and the selling price is 39, then cost price (in) will be
 (1) 20 (2) 22
 (3) 28 (4) 30
- 17.** By selling 1 dozen ball pens, a shopkeeper earned the profit equal to the selling price of 4 ball pens. His profit per cent is
 (1) 50% (2) 40%
 (3) $33\frac{1}{3}\%$ (4) $31\frac{1}{4}\%$
- 18.** A merchant sold an article for 75 at a profit percent equal to its cost price. The cost price of the article was:
 (1) 45 (2) 50
 (3) 54 (4) 60
- 19.** If the profit on selling an article for 425 is the same as the loss on selling it for 355, then the cost price of the article is
 (1) 370 (2) 380
 (3) 390 (4) 400
- 20.** The loss incurred on selling 21 articles equals the selling price of 3 articles. Then the loss per cent is
 (1) $9\frac{1}{11}\%$ (2) 10%
 (3) $12\frac{1}{2}\%$ (4) $11\frac{1}{9}\%$
- 21.** A man sold 250 chairs and had a gain equal to selling price of 50 chairs. His profit per cent is:
 (1) 20% (2) 25%
 (3) 50% (4) 15%
- 22.** On selling 17 balls at 720, there is a loss equal to the cost price of 5 balls. The cost price (in) of a ball is
 (1) 45 (2) 50
 (3) 55 (4) 60
- 23.** A dishonest dealer professes to sell his goods at cost price but uses a weight of 875 gms for the kilogram weight. His gain in percentage is a
 (1) 17% (2) $14\frac{5}{7}\%$
 (3) $14\frac{2}{7}\%$ (4) 14%
- 24.** A dishonest dealer professes to sell his goods at cost price but uses a weight of 800 gms for the kilogram weight. His gain in percentage is a
 (1) 20% (2) 25%
 (3) 30% (4) 35%
- 25.** A dishonest shopkeeper professes to sell goods at his cost price but uses a false weight of 50 gms, for each kilogram. His gain per cent is:
 (1) $6\frac{1}{4}\%$ (2) $5\frac{5}{19}\%$
 (3) $5\frac{3}{17}\%$ (4) $6\frac{2}{7}\%$
- 26.** A dishonest dealer defrauds to the extent of $x\%$ in buying as well as selling his goods by using faulty weight. What will be the gain per cent on his outlay?
 (1) $2x\%$ (2) $\left(\frac{10}{x} + x^2\right)\%$
 (3) $\left(\frac{x^2}{100} + x\right)\%$ (4) None of these
- 27.** A shopkeeper sells rice at 10% profit and uses weight 30% less than the actual measure. His gain per cent is
 (1) $57\frac{2}{3}\%$ (2) $57\frac{1}{7}\%$
 (3) $57\frac{2}{5}\%$ (4) $57\frac{3}{7}\%$
- 28.** A shopkeeper sells rice at 9% loss and uses weight 30% less than the actual measure. His gain per cent is

- (1) 30% (2) 25%
(3) 20% (4) 15%

29. A shopkeeper sells rice at 10% profit and uses weight 20% less than the actual measure. His gain per cent is

- (1) $57\frac{2}{3}\%$ (2) $37\frac{1}{2}\%$
(3) 25% (4) $57\frac{3}{7}\%$

30. A shopkeeper promise to sell his goods at x% profit but he uses 20% less weight thus gain

$37\frac{1}{2}\%$. Find x per cent is?

- (1) 10% (2) 25%
(3) 20% (4) 15%

ANSWER KEY

1.	(2)	2.	(3)	3.	(2)	4.	(4)	5.	(2)	6.	(4)	7.	(2)	8.	(1)	9.	(3)	10.	(1)
11.	(2)	12.	(1)	13.	(3)	14.	(2)	15.	(1)	16.	(4)	17.	(1)	18.	(2)	19.	(3)	20.	(3)
21.	(2)	22.	(4)	23.	(3)	24.	(2)	25.	(2)	26.	(4)	27.	(2)	28.	(1)	29.	(2)	30.	(1)

PRACTICE SHEET 4

- Applied to a bill for 1,00,000 the difference between a discount of 40% and two successive discounts of 36% and 4% is :
(1) Nil (2) 1,440
(3) 2,500 (4) 4,000
- Successive discounts of 10% and 30% are equivalent to a single discount of:
(1) 40% (2) 35%
(3) 38% (4) 37%
- The marked price of a watch was 720/-. A man bought the same for 550.80, after getting two successive discounts, the first at 10%. What was the second discount rate?
(1) 12% (2) 14%
(3) 15% (4) 18%
- An item is marked for 240 for sale. If two successive discounts of 10% and 5% are allowed on the sale price, the selling price of the article will be
(1) 205.20 (2) 204
(3) 34.80 (4) 36
- The price of an article is raised by 30% and then two successive discounts of 10% each are allowed. Ultimately the price of the article is
(1) increased by 10% (2) increased by 5.3%
(3) decreased by 3% (4) decreased by 5.3%
- A single discount equivalent to the successive discounts of 10%, 20% and 25% is
(1) 55% (2) 45%
(3) 46% (4) 60%
- A shopkeeper gives two successive discounts on an article marked 450. The first discount given is 10 per cent. If the customer pays 344.25 for the article, the second discount given is
(1) 14 per cent (2) 10 per cent
(3) 12 per cent (4) 15 per cent
- A company offers three types of successive discounts: (i) 25% and 15%, (ii) 30% and 10%, (iii) 35% and 5%. Which offer is the best for a customer?
(1) First offer
(2) Second offer
(3) Third offer
(4) Any one; all are equally good
- An article is listed at 900 and two successive discounts of 8% and 8% are given on it. How much would the seller gain or lose, if he

gives a single discount of 16%, instead of two discounts?

- (1) Gain of 4.76 (2) Loss of 5.76
(3) Gain of 5.76 (4) Loss of 4.76

- 10.** The marked price of a T.V. is 16,000. After two successive discounts it is sold for 11,400.

If the first discount is 5%, then the rate of second discount is

- (1) 15% (2) 20%
(3) 30% (4) 25%

- 11.** The difference between successive discounts of 40% followed by 30% and 45% followed by 20% on the marked price of an article is Rs.

12. The marked price of the article is:

- (1) 800 (2) 400
(3) 200 (4) 600

- 12.** Buy 4 Get 1 free is equivalent to a discount of

- (1) 15% (2) 20%
(3) 30% (4) 25%

- 13.** Buy 7 Get 3 free is equivalent to a discount of

- (1) 15% (2) 20%
(3) 30% (4) 25%

- 14.** Buy 3 Get 1 free + 20% more discount is equivalent to a discount of

- (1) 25% (2) 30%
(3) 35% (4) 40%

- 15.** Buy 4 Get 1 free + 10% more discount is equivalent to a discount of

- (1) 22% (2) 24%
(3) 26% (4) 28%

- 16.** A trader marked the selling price of an article at 10% above the cost price. At the time of selling, he allows certain discount and suffers a loss of 1%. He allowed the discount of:

- (1) 11% (2) 10%
(3) 9% (4) 10.5%

- 17.** A shopkeeper marks his goods at 30% above the cost price but allows a discount of 10% at the time of sale. His gain is

- (1) 21% (2) 20%
(3) 18% (4) 17%

- 18.** A shopkeeper marks the price of an item keeping 20% profit. If he offers a discount of

$12\frac{1}{2}\%$ on the marked price, his gain

percent will be

- (1) 4.5% (2) 5%
(3) 7.5% (4) 8%

- 19.** A seller marks his goods 30% above their cost price but allows 15% discount for cash payment. His percentage of profit when sold in cash is

- (1) 10.5% (2) 15%
(3) 9% (4) 8.5%

- 20.** A tradesman marks his goods at 25% above its cost price and allows purchasers a discount of $12\frac{1}{2}\%$ for cash payment. The profit, he thus makes, is

- (1) $9\frac{3}{8}\%$ (2) $9\frac{1}{2}\%$
(3) $8\frac{1}{2}\%$ (4) $8\frac{3}{8}\%$

- 21.** What price should a shopkeeper mark on an article costing him 200 to gain 35% after allowing a discount of 25%?

- (1) 270 (2) 300
(3) 330 (4) 360

- 22.** A trader marks his goods 40% above cost price and allows a discount of 25%. The profit he makes, is:

- (1) 15% (2) 10%
(3) 5% (4) 2%

- 23.** A dealer marks his goods 20% above their cost price. He then allows some discount on marked price so that he makes a profit of 10%. The rate of discount is

- (1) $10\frac{1}{3}\%$ (2) $9\frac{1}{3}\%$
(3) $8\frac{2}{3}\%$ (4) $8\frac{1}{3}\%$

- 24.** In a shop, shirts are usually sold at 40% above the cost price. During a sale, the shopkeeper offers a discount of 10% off the usual selling price. If he manages to sell 72 shirts for 13,608, then his cost price per shirt, (in) is

- (1) 210 (2) 150
(3) 149 (4) 125

- 25.** If a shopkeeper marks the price of goods 50% more than their cost price and allows a discount of 40%, what is his gain or loss percent?
 (1) Gain of 10% (2) Loss of 10%
 (3) Gain of 20% (4) Loss of 20%
- 26.** The marked price of a shirt and trousers are in the ratio 1 : 2. The shopkeeper gives 40% discount on the shirt. If the total discount on the set of the shirt and trousers is 30%, the discount offered on the trousers is
 (1) 15% (2) 20%
 (3) 25% (4) 30%
- 27.** A retailer purchases a grinder at a discount of 15% and sells it for 1955 at a profit of 15%. The amount of discount received by the retailer from the wholesaler was
 (1) 270 (2) 290
 (3) 300 (4) 330
- 28.** A dozen pair of socks quoted at 80 are available at a discount of 10%. How many pairs of socks can be bought for 24?
 (1) 4 (2) 5
 (3) 3 (4) 6
- 29.** If an electricity bill is paid before due date, one gets a reduction of 4% on the amount of the bill. By paying the bill before due date a person got a reduction of 13. The amount of his electricity bill was
 (1) 125 (2) 225
 (3) 325 (4) 425
- 30.** A shopkeeper lists the price of an article as 500. But he gives a certain discount which allows the buyer to pay 500 for the article including 10% sales tax. The rate of discount is
 (1) 10% (2) $10\frac{1}{11}\%$
 (3) $9\frac{1}{11}\%$ (4) 11%.

ANSWER KEY

1.	(2)	2.	(4)	3.	(3)	4.	(1)	5.	(2)	6.	(3)	7.	(4)	8.	(3)	9.	(2)	10.	(4)
11.	(4)	12.	(2)	13.	(3)	14.	(4)	15.	(4)	16.	(2)	17.	(4)	18.	(2)	19.	(1)	20.	(1)
21.	(4)	22.	(3)	23.	(4)	24.	(2)	25.	(2)	26.	(3)	27.	(3)	28.	(1)	29.	(3)	30.	(3)

PRACTICE SHEET 5

- 1.** A person sells two machines at 396 each. On one he gains 10% and on the other he loses 10%. His profit or loss in the whole transaction is :
 (1) no gain no loss (2) 1% loss
 (3) 1% profit (4) 8% profit
- 2.** A dealer sold two types of goods for 10,000 each. On one of them, he lost 20% and on the other he gained 20%. His gain or loss per cent in the entire transaction was
 (1) 2% loss (2) 2% gain
 (3) 4% gain (4) 4% loss
- 3.** A television and a refrigerator were sold for 12,000 each. If the television was sold at a loss of 20% of the cost and the refrigerator at a gain of 20% of the cost, the entire transaction resulted in
 (1) No loss or gain (2) Loss of 1,000
 (3) Gain of 1,000 (4) Loss of 1,200
- 4.** A man had 100 kgs of sugar, part of which he sold at 7% profit and rest at 17% profit. He gained 10% on the whole. How much did he sell at 7% profit?
 (1) 65 kg (2) 35 kg
 (3) 30 kg (4) 70 kg
- 5.** A man bought two goats for 1008. He sold one at a loss of 20% and other at a profit of 44%. If each goat was sold for the same price, the cost price of the goat which was sold at loss, was:
 (1) 648 (2) 360
 (3) 568 (4) 440

6. Two bicycles were sold for 3990 each, gaining 5% on one and losing 5% on other. The gain or loss per cent on the whole transaction is :
 (1) neither gain nor loss
 (2) 2.5% gain
 (3) 2.5% loss
 (4) 0.25% loss
7. A man sold two watches for 240 each. On one he gains 20% and incurs a loss of 20% on another. What is his gain or loss per cent in this transaction?
 (1) 1% profit (2) 2% loss
 (3) 4% profit (4) 4% loss
8. When the price of cloth was reduced by 25%, the quantity of cloth sold increased by 20%. What was the effect on gross receipt of the shop?
 (1) 5% increase (2) 5% decrease
 (3) 10% increase (4) 10% decrease
9. A cloth merchant sold half of his cloth at 20% profit, half of the remaining cloth at 20% loss and the rest was sold at his cost price. In the total transaction, his gain or loss will be
 (1) 5% profit
 (2) Neither loss nor gain
 (3) 5% loss
 (4) 10% profit
10. The total cost price of two watches is 840. One is sold at a profit of 16 per cent and the other at a loss of 12 per cent. There is no loss or gain in the whole transaction. The cost price of the watch on which the shopkeeper gains, is
 (1) 360 (2) 370
 (3) 380 (4) 390
11. A shopkeeper purchased 510 eggs at the rate of Rs. 20 per dozen. 30 eggs were broken on the way. In order to make a gain of 20%, he must sell the remaining eggs at the rate of
 (1) Rs. 22.50 per dozen
 (2) Rs. 25.50 per dozen
 (3) Rs. 26 per dozen
 (4) Rs. 26.50 per dozen
12. A sells a watch to B and makes a loss of 12%. B makes a profit of $12\frac{1}{2}\%$ by selling the watch to C. If A sells the watch to B at the cost of which C purchased it, then the percentage of loss or profit of A will be
 (1) 1% loss (2) 1% profit
 (3) 2% loss (4) 2% profit
13. A man buys 3 type-I cakes and 6 type-II cakes for Rs. 900. He sells type-I cakes at a profit of 15% and type-II cakes at a loss of 10%. If his overall profit is Rs.30, the cost price (in Rs.) of a type-I and of a type-II cakes is
 (1) 100, 100 (2) 160, 70
 (3) 180, 60 (4) 120, 90
14. A merchant buys 25 litres of milk daily at the rate of Rs. 12 per litre. He mixes 5 litres of water in it and sells at the rate Rs. 10.40 per litre. His gain is :
 (1) 8% profit (2) 2% profit
 (3) 4% profit (4) 6% profit
15. A trader had 22 quintals of wheat. He sold a part of it at 23% profit and the rest at 33% profit, so that he made a total profit of 27%. How much wheat did he sell at 33% profit?
 (1) 1320 kg. (2) 440 kg.
 (3) 880 kg. (4) 1760 kg.
16. A milkman mixes water with milk and sells the mixture at the cost price of pure milk. The volume of water in litres to be mixed with each litre of milk to get a 25% profit is
 (1) $\frac{1}{4}$
 (2) $\frac{1}{5}$
 (3) $1\frac{1}{4}$
 (4) cannot be calculated without knowing the cost price of milk
17. The difference between the selling prices of an article at a profit of 15% and at a profit of 10% is 10. The cost price of the article is
 (1) 100 (2) 120
 (3) 150 (4) 200
18. The difference between the selling price and cost price of an article is 210. If the profit percent is 25, then the selling price of the article is
 (1) 950 (2) 1,050
 (3) 1,150 (4) 1,250

- 19.** A sells a cycle to B at a profit of 10%, B sells to C at a profit of 20%. If C pays 264 for it, how much did A pay for it?
 (1) 200 (2) 220
 (3) 225 (4) 234
- 20.** A man purchased an article and sold it to B at a profit of 25% and B sold it to C at a loss of 10% and C paid 675 for it. For how much did A purchase it (in)?
 (1) 625 (2) 575
 (3) 600 (4) 550
- 21.** A milkman makes 20% profit by selling milk mixed with water at 9 per litre. If the cost price of 1 litre pure milk is 10, then the ratio of milk and water in the said mixture is
 (1) 3 : 1 (2) 4 : 1
 (3) 3 : 2 (4) 4 : 3
- 22.** Nita blends two varieties of tea-one costing 180 per kg and another costing 200 per kg in the ratio 5 : 3. If she sells the blended variety at 210 per kg, then her gain percent is
 (1) 10% (2) 11%
 (3) 12% (4) 13%
- 23.** In what ratio Darjeeling Tea costing 320 per kg be mixed with Assam Tea costing 250 per kg so that there is a gain of 20% by selling the mixture at 324 per kg ?
 (1) 1 : 2 (2) 2 : 3
 (3) 3 : 2 (4) 2 : 5
- 24.** The ratio, in which tea costing 192 per kg is to be mixed with tea costing 150 per kg so that the mixed tea, when sold for 194.40 per kg, gives a profit of 20%, is
 (1) 2 : 5 (2) 3 : 5
 (3) 5 : 3 (4) 5 : 2
- 25.** The ratio of the quantities of sugar, in which sugar costing 20 per kg. and 15 per kg. should be mixed so that there will be neither loss nor gain on selling the mixed sugar at the rate of 16 per kg, is
 (1) 2 : 1 (2) 1 : 2
 (3) 4 : 1 (4) 1 : 4
- 26.** The ratio in which the Darjeeling tea at 32 per kg is mixed with the Assam tea at 25 per kg so as to gain 20% by selling the mixture at 32.40 per kg is
 (1) 4 : 3 (2) 3 : 4
 (3) 5 : 2 (4) 2 : 5
- 27.** 7 kg of tea costing 280 per kg is mixed with 9 kg of tea costing 240 per kg. The average price per kg of the mixed tea is
 (1) 255.80 (2) 257.50
 (3) 267.20 (4) 267.50
- 28.** A shopkeeper bought 15kg of rice at the rate of 29 per kg and 25kg of rice at the rate of 20 per kg. He sold the mixture of both types of rice at the rate of 27 per kg. His profit in this transaction is
 (1) 125 (2) 150
 (3) 140 (4) 145
- 29.** Costs of two watches were in the ratio of 16 : 23. The cost of first watch increases by 10% and that of second by Rs. 477. Now the costs of two watches are in a ratio of 11 : 20. The price of the second watch (in Rs.) in the beginning was
 (1) 932 (2) 1219
 (3) 1696 (4) 848
- 30.** The liquids, X and Y are mixed in the ratio of 3 : 2 and the mixture is sold at Rs. 11 per litre at a profit of 10%. If the liquid X costs Rs. 2 more per litre than Y, the cost of X per litre is (in Rs.) :
 (1) 10.80 (2) 11.75
 (3) 9.50 (4) 11

ANSWER KEY

1.	(2)	2.	(4)	3.	(2)	4.	(4)	5.	(1)	6.	(4)	7.	(4)	8.	(4)	9.	(1)	10.	(1)
11.	(2)	12.	(1)	13.	(2)	14.	(3)	15.	(3)	16.	(2)	17.	(4)	18.	(2)	19.	(1)	20.	(3)
21.	(1)	22.	(3)	23.	(4)	24.	(1)	25.	(4)	26.	(4)	27.	(2)	28.	(4)	29.	(2)	30.	(1)

CDS PYQ

1. Two lots of onions with equal quantity, one costing 10 per kg and the other costing Rs 15 per kg, are mixed together and whole lot is sold at Rs 15 per kg. What is the profit or loss?
[CDS 2013- II]
(a) 10% loss (b) 10% profit
(c) 20% profit (d) 20% loss
2. A person sold an article from Rs 3600 and got a profit of 20%. Had he sold the article for Rs 3150, how much profit would he have got?
[CDS 2013- II]
(a) 4% (b) 5%
(c) 6% (d) 10%
3. On a 20% discount sale, an article costs 596. What was the original price of the article?
[CDS 2014- I]
(a) Rs 720 (b) Rs 735
(c) Rs 745 (d) Rs 775
4. A person selling an article for 96 finds that his loss per cent is one-fourth of the amount of rupees that he paid for the article, what can be the cost price?
[CDS 2014- II]
(a) Only Rs 160
(b) Only Rs 240
(c) Either Rs 160 or Rs 240
(d) Neither Rs 160 nor Rs 240
5. A shopkeeper sells his articles at their cost price but uses a faulty balance which reads 1000g for 800g. What is his actual profit percentage?
[CDS 2014- II]
(a) 25% (b) 20%
(c) 40% (d) 30%
6. A man buys 200 oranges for R 1000. How many oranges for 100 can be sold, so that his profit percentage 25% is:
[CDS 2014- II]
(a) 10 (b) 14
(c) 16 (d) 20
7. When an article is sold at 20% discount, the selling price is Rs 24. What will be the selling price when the discount is 30%?
[CDS 2014- II]
(a) Rs. 25 (b) Rs 23
(c) Rs 21 (d) Rs 20
8. A milkman claims to sell milk at its cost price only, but he is making a profit of 20% since he has mixed some amount of water in the milk. What is the percentage of milk in the mixture?
[CDS 2015- I]
(a) 80% (b) $\frac{250}{3}\%$
(c) 75% (d) $\frac{200}{3}\%$
9. The value of a single discount on some amount which is equivalent to a series of discounts of 10%, 20% and 40% on the same amount, is equal to
[CDS 2015- II]
(a) 43.2% (b) 50%
(c) 56.8% (d) 70%
10. A milkman claims to sell milk at its cost price only. Still he is making a profit of 20% since he has mixed some amount of water in the milk. What is percentage of milk in the mixture?
[CDS 2015- II]
(a) $200/3\%$ (b) 75%
(c) 80% (d) $250/3\%$
11. A cloth merchant buys cloth from a weaver and cheats him by using a scale which is 10 cm longer than a normal metre scale. He claims to sell cloth at the cost price to his customers, but while selling uses a scale which is 10 cm shorter than a normal metre scale. What are his gains?
[CDS 2016- I]
(a) 20% (b) 21%
(c) $22\frac{2}{9}\%$ (d) 23%

- 12.** The cost of 2.5 kg rice is Rs 125. The cost of 9 kg rice is equal to that of 4 kg pulses. The cost of 14 kg pulses is equal to that of 1.5 kg tea. The cost of 2 kg tea is equal to that of 5 kg nuts. What is the cost of 11 kg nuts?

[CDS 2016- II]

- (a) Rs 2310 (b) Rs 3190
(c) Rs 4070 (d) Rs 4620

- 13.** A shopkeeper increases the cost price of an item by 20% and offers a discount of 10% on this marked price. What is his percentage gain?

[CDS 2016- II]

- (a) 15% (b) 12%
(c) 10% (d) 8%

- 14.** Rajendra bought a mobile with 25% discount on the selling price. If the mobile cost him 4,875, what is the original selling price of the mobile?

[CDS 2017- I]

- (a) Rs 6,300 (b) Rs 6,400
(c) Rs 6,500 (d) Rs 6,600

- 15.** Sudhir purchased a chair with three consecutive discounts of 20%, 12.5% and 5%. The actual deduction will be

[CDS 2017- II]

- (a) 33.5% (b) 30%
(c) 32% (d) 35%

- 16.** Ram buys 4 chairs and 9 stools for Rs 1,340. If he sells chairs at 10% profit and stools at 20% profit, he earns a total profit of Rs 188. How much money did he have to pay for the chairs?

[CDS 2017- II]

- (a) Rs 200 (b) Rs 400
(c) Rs 800 (d) Rs 1,600

- 17.** An article is sold at a profit at 32%. If the cost price is increased by 20% and the sale price remains the same, then the profit percentage becomes

[CDS 2018- I]

- (a) 10% (b) 12%
(c) 15% (d) 20%

- 18.** A milk vendor bought 28 litres of milk at the rate of Rs 8.50 per litre. After adding some water he sold the mixture at the same price.

If his gain is 12.5%, how much water did he add?

[CDS 2018- II]

- (a) 4.5 litres (b) 4 litres
(c) 3.5 litres (d) 3 litres

- 19.** If an article is at a gain of 6% instead of a loss of 6%, the seller gets Rs 6 more. What is the cost price of the article?

[CDS 2019- I]

- (a) Rs 18 (b) Rs 36
(c) Rs 42 (d) Rs 50

- 20.** A trader sells two computers at the same price, making a profit of 30% on one and a loss of 30% on the other. What is the net loss or profit percentage on the transaction?

[CDS 2019- II]

- (a) 6% loss (b) 6 % gain
(c) 9 % loss (d) 9 % gain

- 21.** A person sells two items each at Rs 990, one at a profit of 10% and another at a loss of 10%. What is the combined percentage of profit or loss for two items?

[CDS 2019- II]

- (a) 1% loss (b) 1% profit
(c) No profit no loss (d) 0.5% profit

- 22.** A shopkeeper sells his articles at their cost price but uses a faulty balance which reads 1000gm for 800gm. What is the actual profit percentage?

[CDS 2020- II]

- (a) 20% (b) 25%
(c) 30% (d) 40%

- 23.** If a television set is sold at Rs x, a loss of 28% would be incurred. If it is sold at Rs y, a profit of 12% would be incurred. What is the ratio of y to x?

[CDS 2020- II]

- (a) 41 : 9 (b) 31 : 9
(c) 23 : 9 (d) 14 : 9

- 24.** A person sold an article for Rs 75 which cost him Rs x. He finds that he realized x % profit on his outlay. What is x equal to?

[CDS 2020- II]

- (a) 20 (b) 25
(c) 50 (d) 100

25. A trader gives successive discounts of 20%, 10% and 5% respectively. What is the overall discount?

[CDS 2021- I]

- (a) 30% (b) 31.6%
(c) 32.5% (d) 35%

26. The cost price of 100 mangoes is equal to the selling price of 80 mangoes. What is the profit percentage?

[CDS 2021- I]

- (a) 16% (b) 20%
(c) 24% (d) 25%

27. A person bought chair and a table for Rs 750. He sold the chair at a gain of 5% and the table at a gain of 20%. He gained 16% of the whole. What is the original cost of the table?

[CDS 2021- II]

- (a) Rs 400 (b) Rs 450
(c) Rs 550 (d) Rs 600

28. A shopkeeper marks the price of an article at Rs 200. After allowing a discount of 10%, he still gains 20% on the cost price. What is the cost price of the article?

[CDS 2022 (I)]

- (a) Rs 170 (b) Rs 160
(c) Rs 150 (d) Rs 120

29. Twelve percent of bananas bought by a fruit vendor got lost during transportation. On

selling the remaining bananas, the vendor's overall profit turned out to be 4%. If the vendor had not lost any bananas and had sold them at the price of the remaining bananas, what would have been his profit percentage?

[CDS - 2023 (1)]

- (a) $8\frac{1}{9}\%$ (b) $9\frac{1}{11}\%$
(c) $18\frac{2}{11}\%$ (d) None of the above

30. A person bought an article and sold it at a profit of 20%. Had he bought it at 20% less what would have been the profit percentage if the selling price had been the same?

[CDS-2023 (2)]

- (a) 25% (b) 40%
(c) 50% (d) 60%

31. The cost price of y articles is equal to selling price of z articles. If $y : z = 5 : 4$, what is the profit percentage?

[CDS-2024 (1)]

- (a) 20% (b) 25%
(c) 30% (d) 40%

32. A person bought a book at $\frac{3}{4}$ th of its listed price and sold it at 50% more than its listed price. What is the percentage of gain in the transaction?

[CDS-2024 (1)]

- (a) 20% (b) 40%
(c) 75% (d) 100%

ANSWER KEY

1.	(c)	2.	(b)	3.	(c)	4.	(c)	5.	(a)	6.	(c)	7.	(c)	8.	(b)	9.	(c)	10.	(d)
11.	(c)	12.	(d)	13.	(d)	14.	(c)	15.	(a)	16.	(c)	17.	(a)	18.	(c)	19.	(d)	20.	(c)
21.	(a)	22.	(b)	23.	(d)	24.	(c)	25.	(b)	26.	(d)	27.	(c)	28.	(c)	29.	(c)	30.	(c)
31.	(b)	32.	(d)																

CAPF PYQ

1. If 9 mangoes cost as much as 5 oranges, 5 oranges as much as 3 apples, 4 apples as much as 9 pineapples and if 3 pineapples cost Rs 48, what will a mango cost?
[CAPF 2013]
(a) Rs 9 (b) Rs 12
(c) Rs 18 (d) Rs 27
2. By selling an article of Rs 270 a man loses 10%. If he would sell it at Rs 360 his gain percent is
[CAPF 2013]
(a) 10 (b) 15
(c) 20 (d) 25
3. A, B' and 'C' started their independence businesses with equal amounts of capital. During the first year 'A' made 10% profit, 'B' incurred 10% loss and 'C' made a profit of 5%. In the second year 'A' incurred of 20% loss, B' made profit of 20% and 'C' made profit of 5%. Which of the following is FALSE at the end of second year?
[CAPF 2015]
(a) 'C' is the richest (b) A' is the poorest
(c) 'B' is the richest (d) 'C' is richer than 'B'
4. By selling an article for Rs 2700, a man loses 10%. If he sells it for Rs 3600, his gain per cent is
[CAPF 2019]
(a) 15 (b) 18
(c) 20 (d) 25
5. The cost of gold varies directly as the cube of its weight. A gold piece weighing 20 decigram costs 1,000. If it is broken into two pieces whose weights are in the ratio 2:3, then what is the profit or loss incurred?
[CAPF 2020]
(a) 280 profit (b) 280 loss
(c) 720 profit (d) 720 loss
6. In a stock clearance sell, a shopkeeper gives 40% off on all his items. He incurs a loss of 25% by selling by selling a item of marked price Rs 2000, Which one of the following is the cost price of item?
[CAPF 2021]
(a) Rs 1800 (b) Rs 1200
(c) Rs 1600 (d) Rs 1400
7. A shopkeeper gives two consecutive discounts of 10% and 5% respectively on his items, He then add 20% GST on his items. If an item has marked price Rs 2000, how much more or less of the actual price of the item a customer has to pay?
[CAPF 2021]
(a) 2.6% less (b) 2.6% more
(c) Same price (d) 5.2% more
8. A person buys an item from a shop for which the shopkeeper offers a discount of 10% on the market price. The person pays using an e-wallet which gives 10% cash back. Which one of the following is the value of effective discount?
[CAPF 2022]
(a) 20% (b) 18%
(c) 19% (d) 21%

ANSWER KEY

1.	(b)	2.	(c)	3.	(c)	4.	(c)	5.	(d)	6.	(c)	7.	(b)	8.	(c)				
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